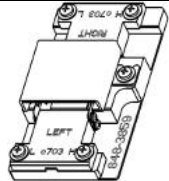


Ultra Sonic Mixer adjustment

1. Adjustment tool	2
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1. Adjustment tool

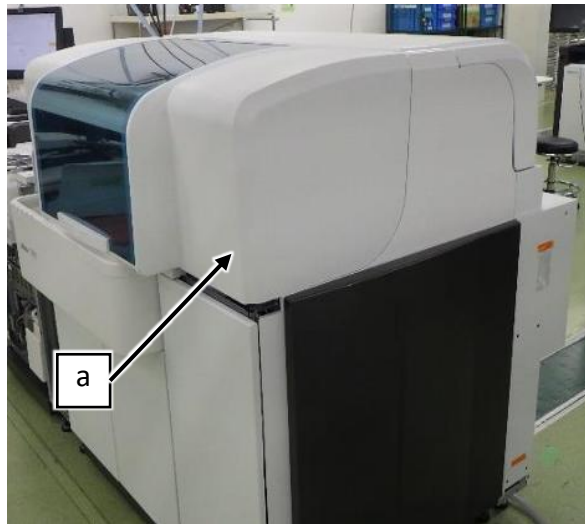
The following adjustment tools are used for this adjustment. Please refer to "S03-01 List of adjustment tools" for a list of jigs.

P/N	Name	Appearance
848-3859	USM HEIGHT TOOL 2	
848-3946	USM TOOL HORIZONTAL 2	

2. Preparation

2.1 Removing the covers

- (1) Open COVER TOP F ASSY (a).



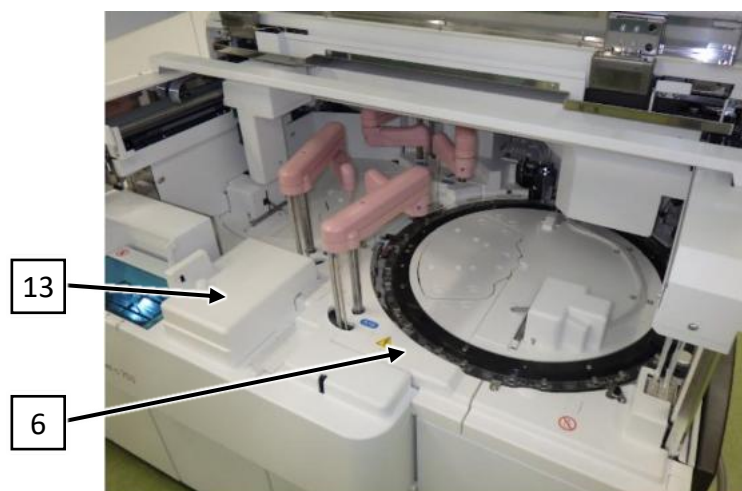
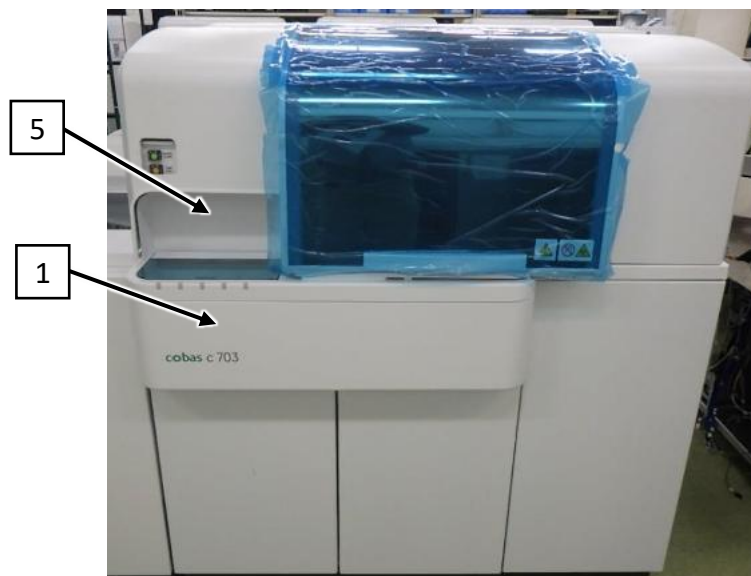
- (2) Loosen the screw and remove COVER MSW ASSY (a).

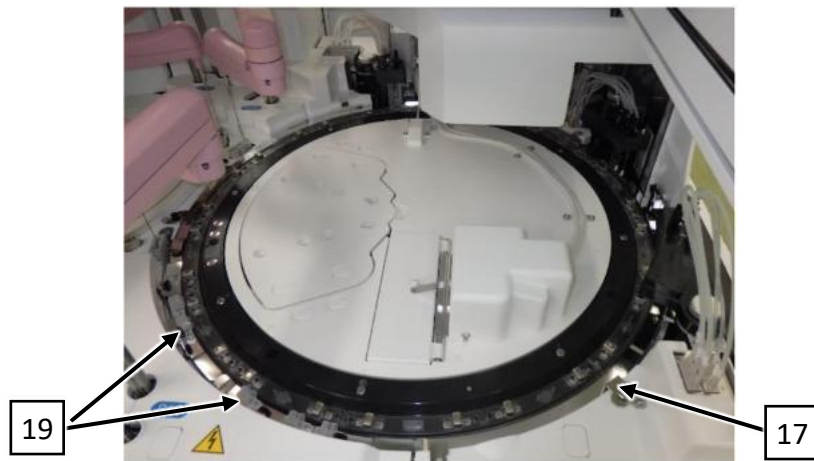


- (3) Turn on Maintenance switch.



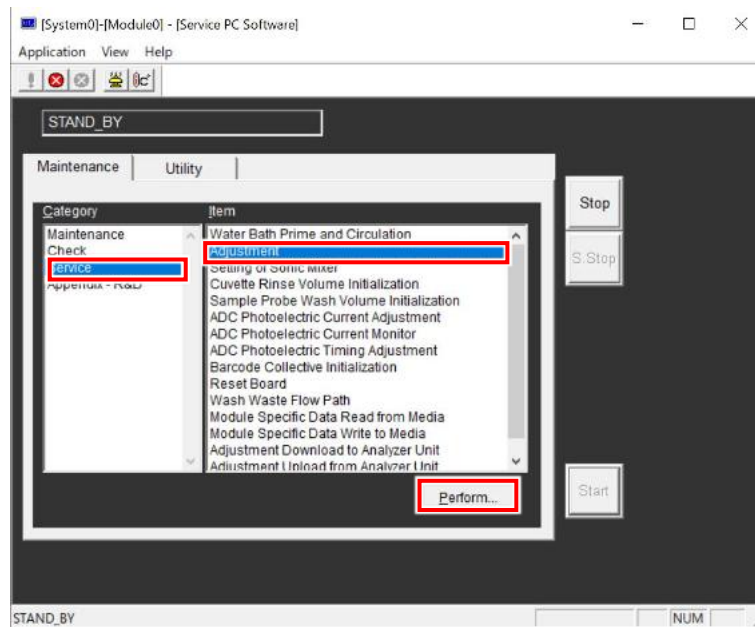
- (4) Remove No.1, No. 5, No. 6, No. 13, No.17 and No.19 covers (refer to "S06-41 Removal and attachment of covers").





2.2 Start SSW

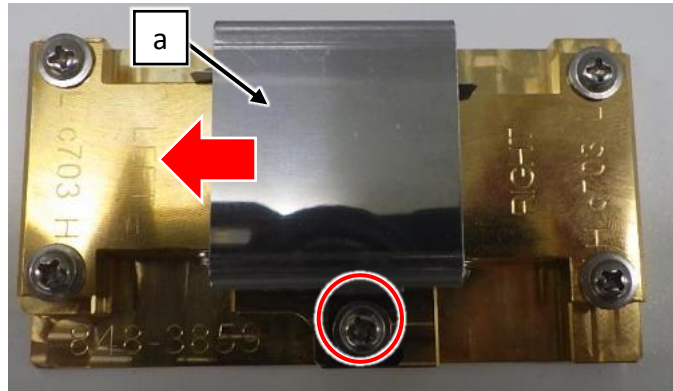
- (1) Refer to “17 pro v2 SSW Adjustment Procedure”.
- (2) Select [Service] > [Adjustment] > [Perform...].



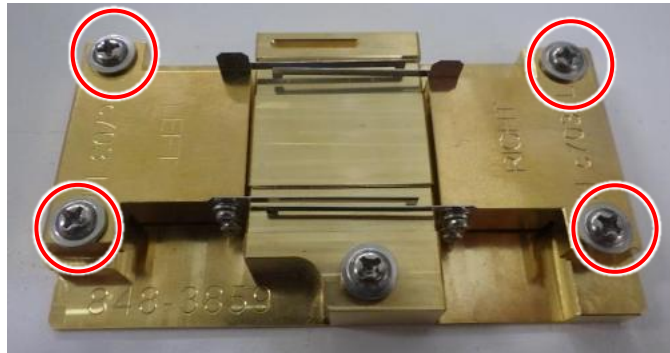
3. Adjustment of Ultrasonic Mixer Vertical

3.1 Checking the arms

- (1) Loosen the screw and remove the cover (a).



- (2) Make sure that tighten the 4 screws.

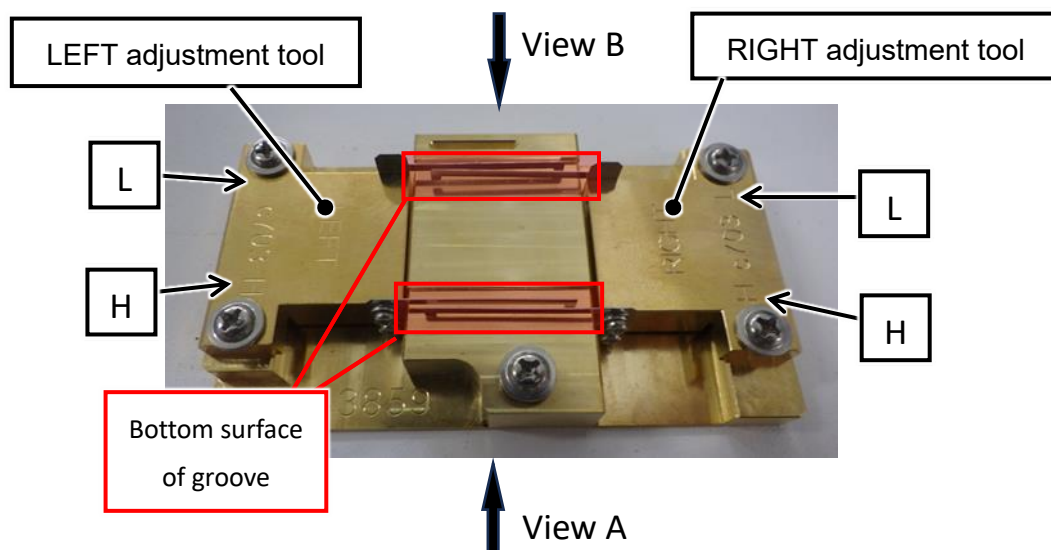


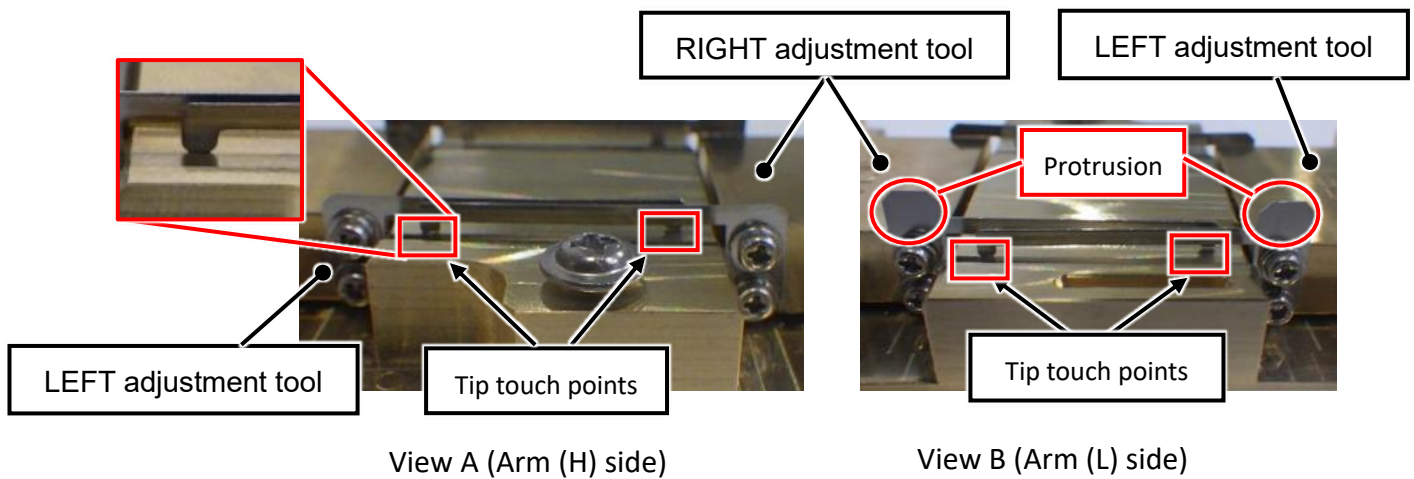
- (3) Make sure that each arm tips touch the bottom surface of groove.

Note that the arm (L) has a protrusion at the base.

If the 4 tips touch the bottom surface, proceed to the step chapter 3.3.

If not, proceed to the next chapter.

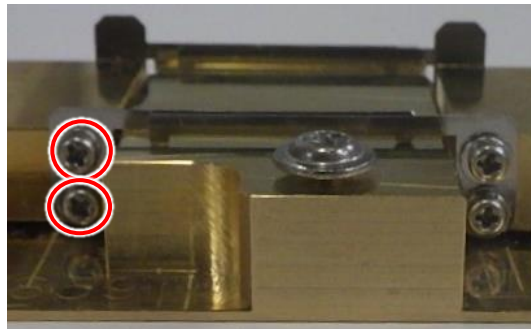




3.2 Adjusting the arms

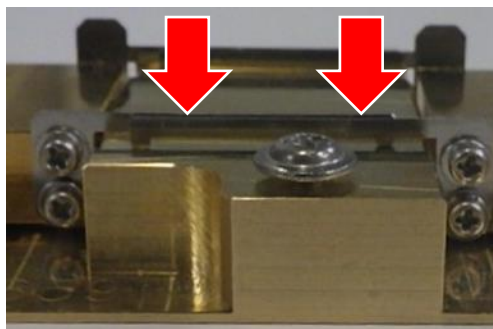
The adjustment procedure for Arm (H) of LEFT adjustment tool, other arms are basically same as the procedure for it. Make sure that the tip of the arm (H) of LEFT adjustment tool touches the bottom surface of groove.

- (1) Loosen the 2 screws for Arm (H) of LEFT adjustment tool.



View A (Arm (H) of LEFT adjustment tool)

- (2) Push the arms lightly so that its tip touches the bottom surface of groove.
Note that tip height changes when each 2 screws are loosened and when each 2 screws are tightened.
Note that pressing the arms with strong force will deform them.

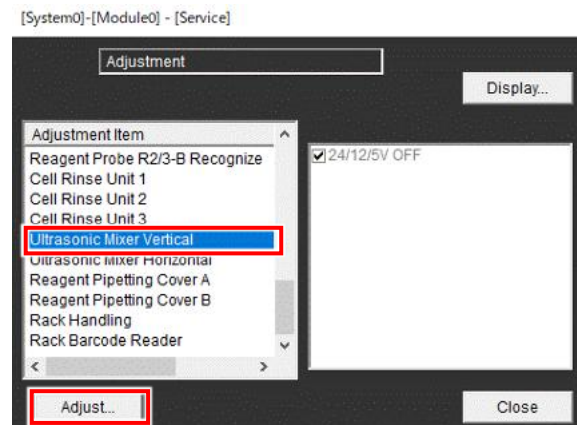


View A (Arm (H) of LEFT adjustment tool)

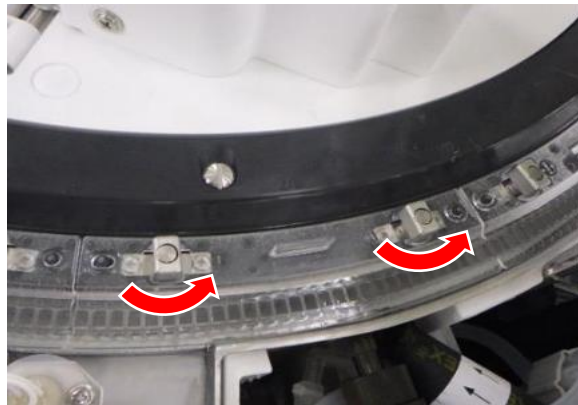
- (3) Tighten the 2 screws.
- (4) Make sure that its tips touch the bottom surface of groove again.
Check on all arms. Adjust the tips of arm, if necessary.
If all tips touch the bottom surface, proceed to the chapter 3.3.
If not, retry from the step (1).

3.3 Adjustment of Ultrasonic Mixer Vertical

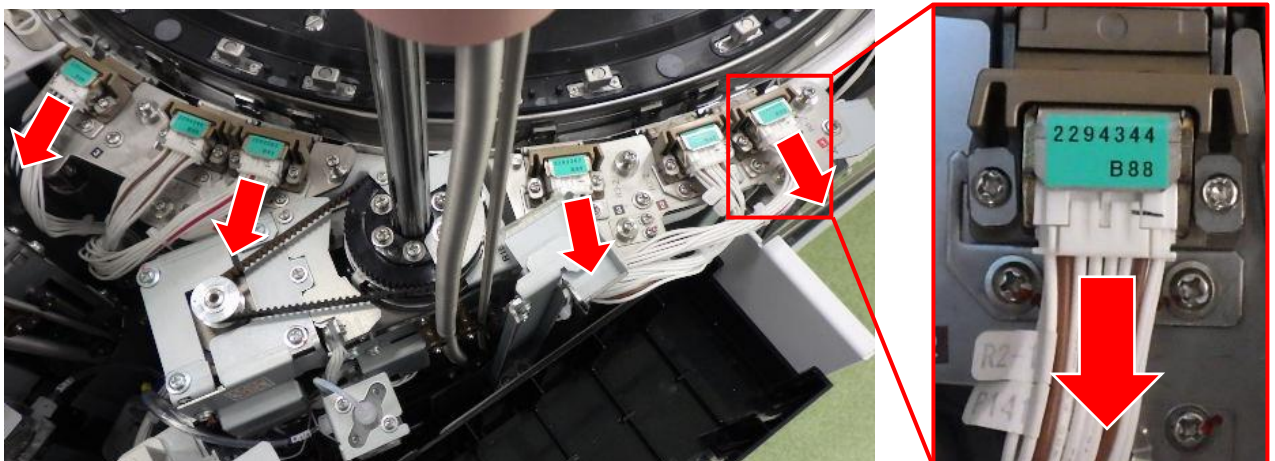
- (1) Select [Ultrasonic Mixer Vertical] from the Adjustment Item and select [Adjust...].



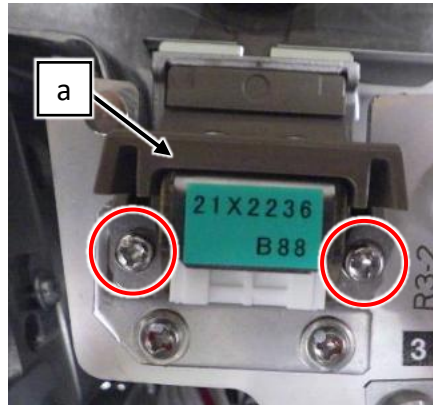
- (2) Twist the 2 quick screws and remove the cell. Similarly remove all the other cells.



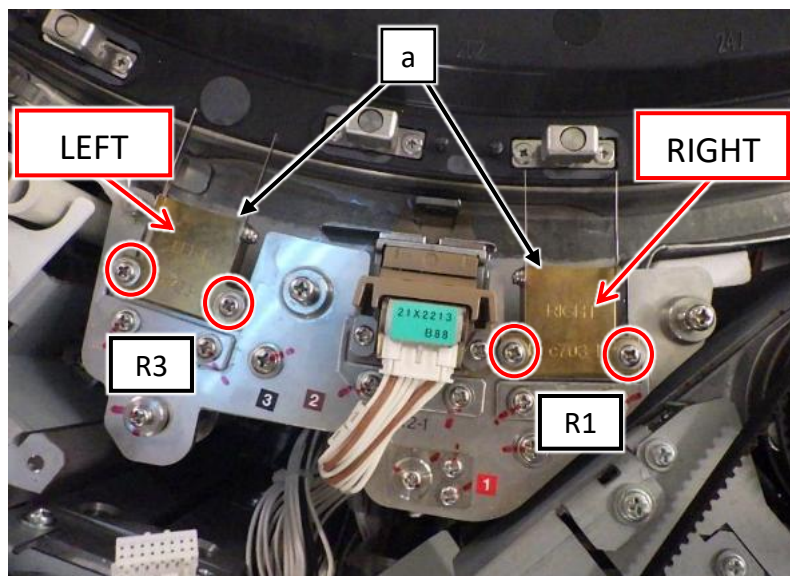
- (3) Remove the 4 connectors.



- (4) Remove the 2 screws and USM unit (a). Similarly remove all the other USM units.



- (5) Set the tools (a) with the letter "LEFT" at the position of R3 and the tool (a) with the letter "RIGHT" at the position of R1 and tighten the 4 screws.

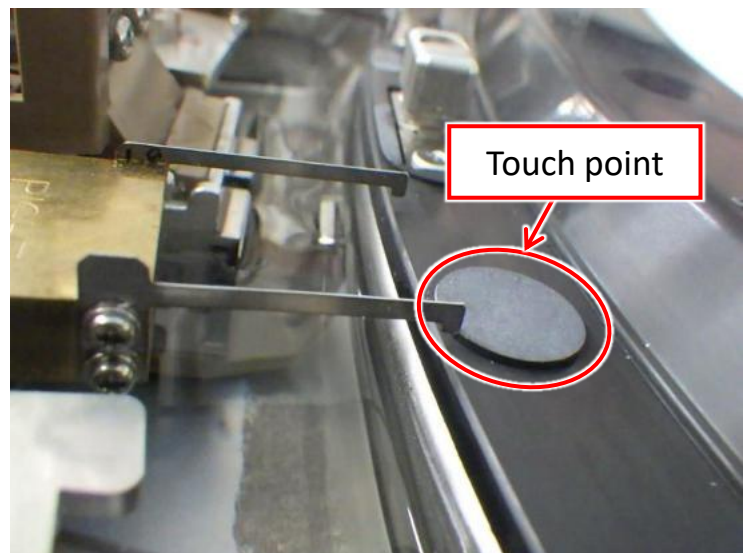
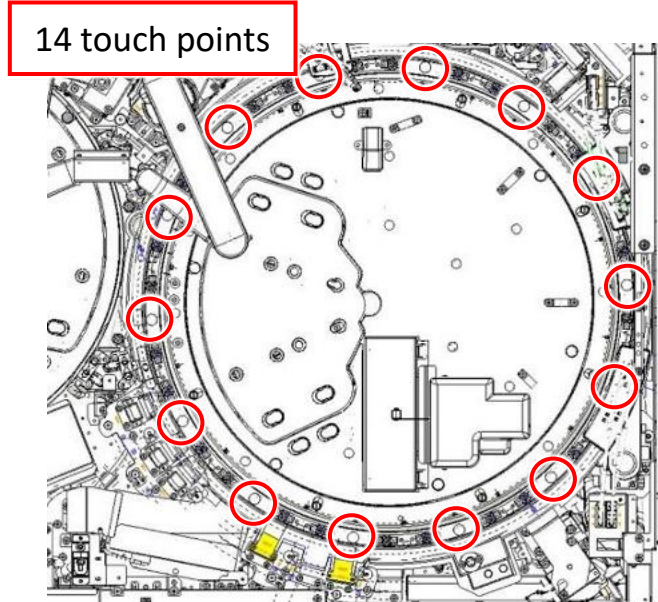
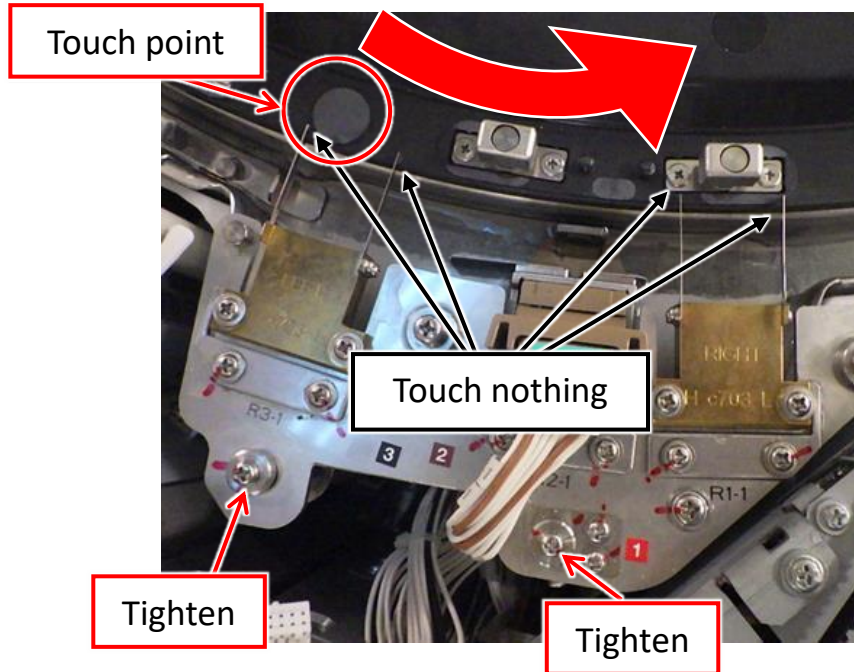


- (6) Rotate Reaction disk counterclockwise and make sure that the 2 arm (H) tips and the 2 arm (L) tips do not touch the touch points.

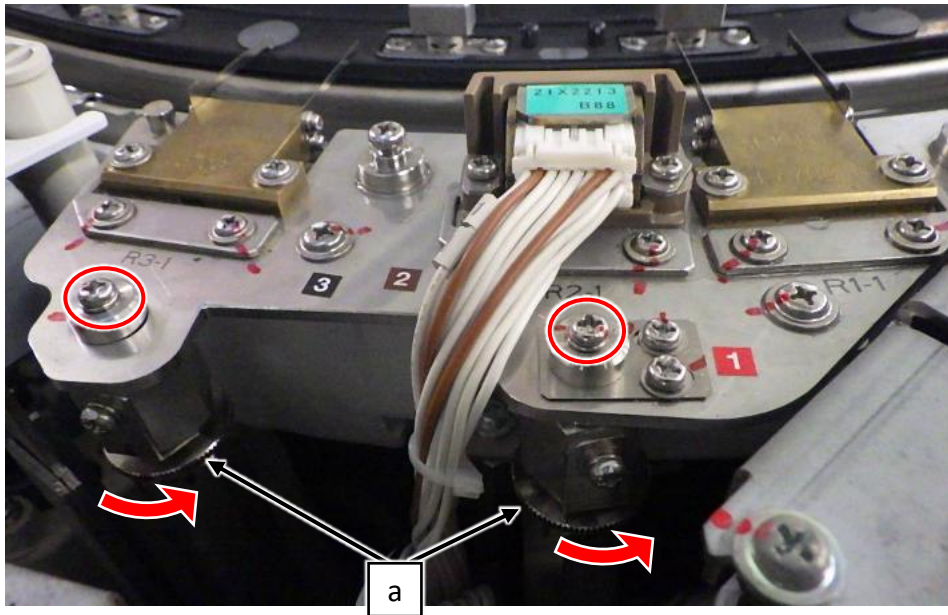
Note: the tip height changes when the 2 screws are loosened and when the 2 screws are tightened.

If they touch nothing, proceed to the step (11).

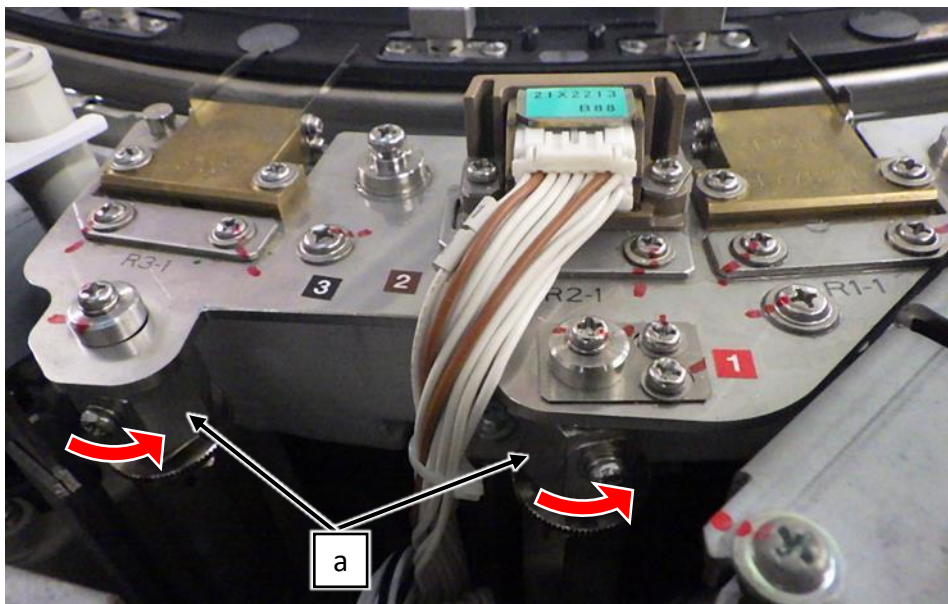
If they touch something, proceed to the step (7).



- (7) Loosen the 2 screws and the 2 lock nuts (a) counterclockwise.



- (8) Rotate the 2 hexagonal parts (a) counterclockwise to lift the tools.

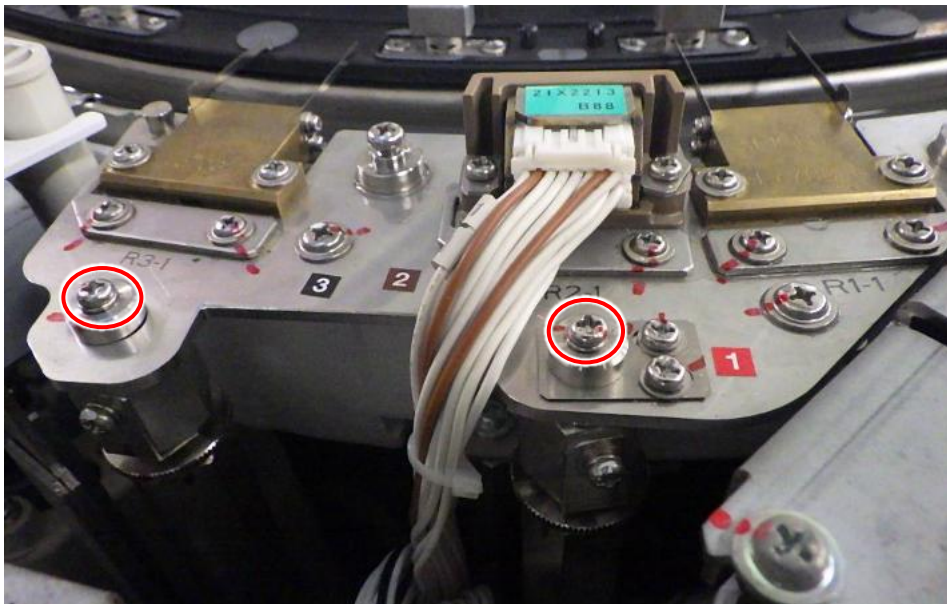


(9) Tighten the 2 screws.



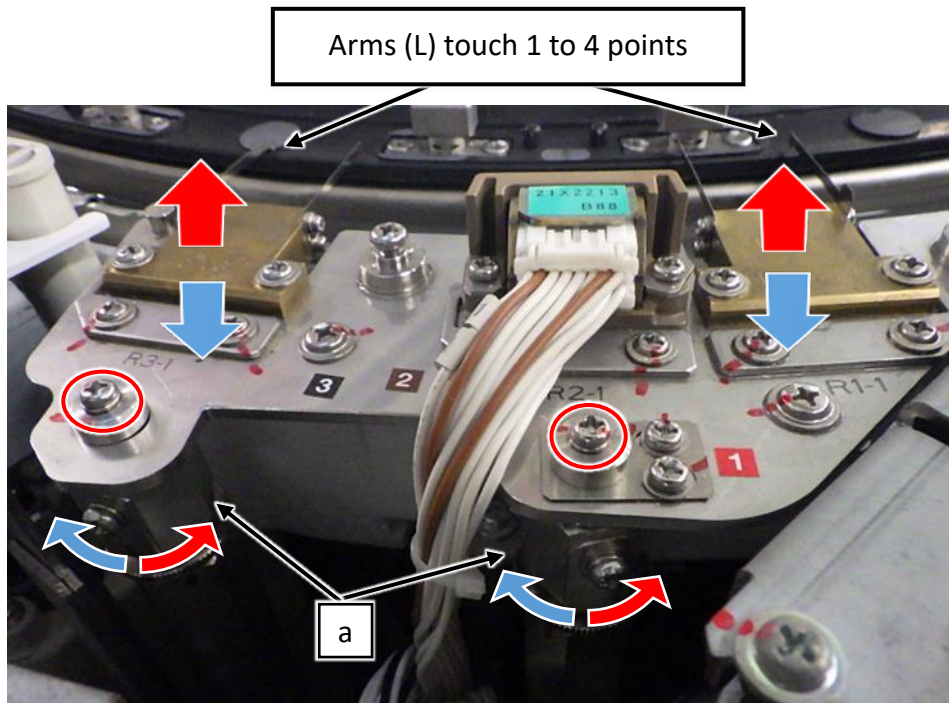
(10) Retry from the step (6).

(11) Loosen the 2 screws.

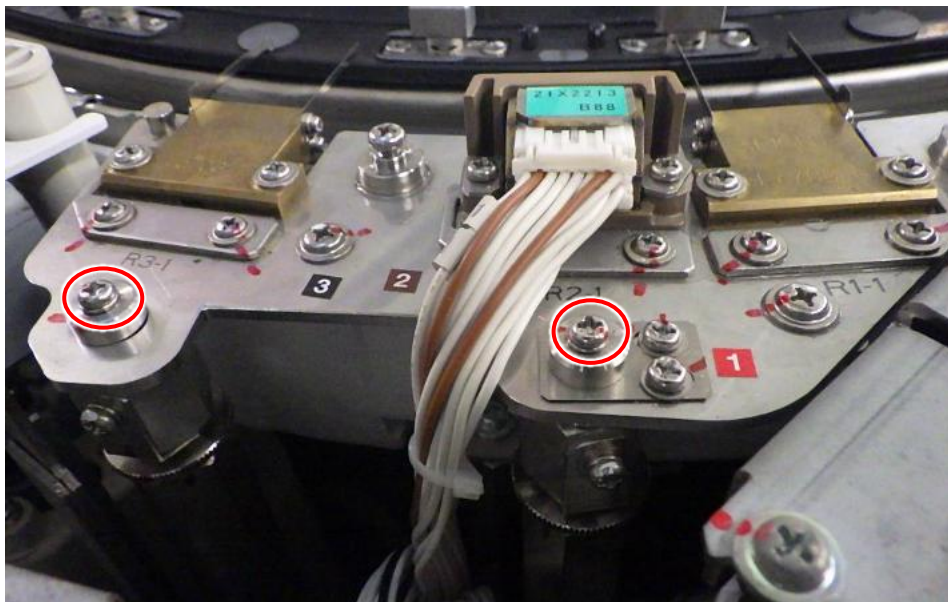


- (12) Rotate the 2 hexagonal parts (a) to make each arm (L) tip touch the touch point 1 to 4 when rotate Reaction disk once counterclockwise and the 2 screws are tightened.

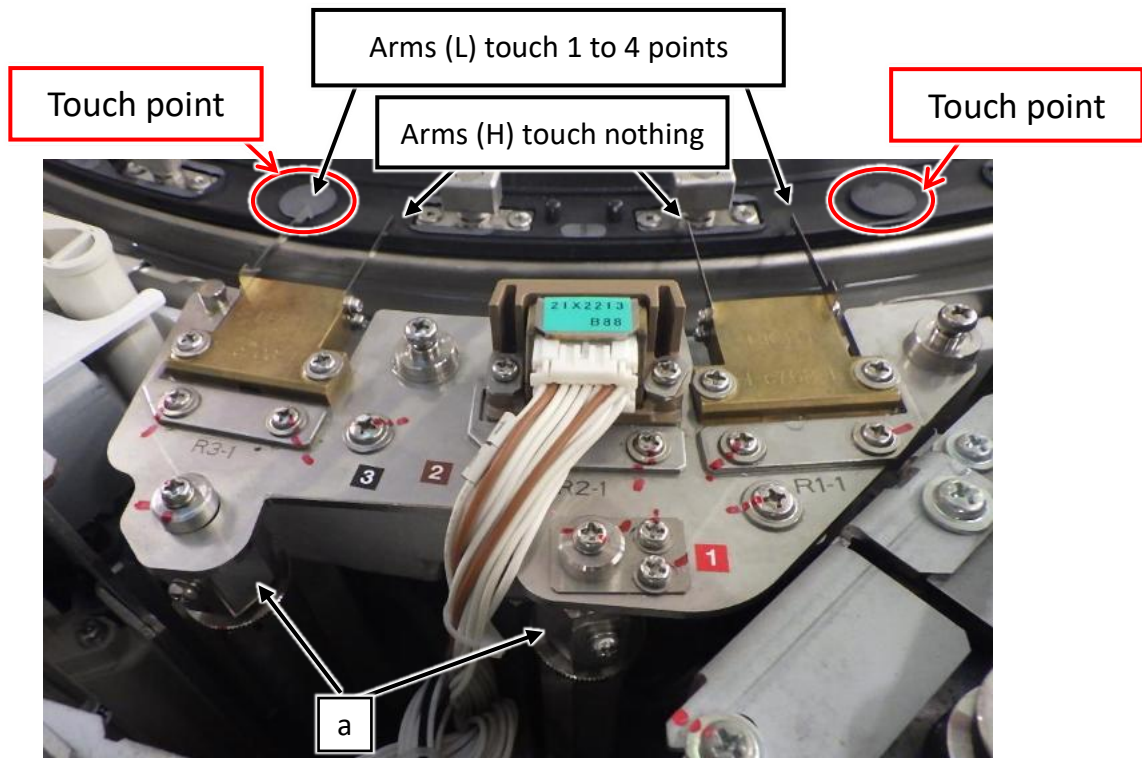
Note that the tip height changes when the 2 screws are loosened and when the 2 screws are tightened.



- (13) Tighten the 2 screws.

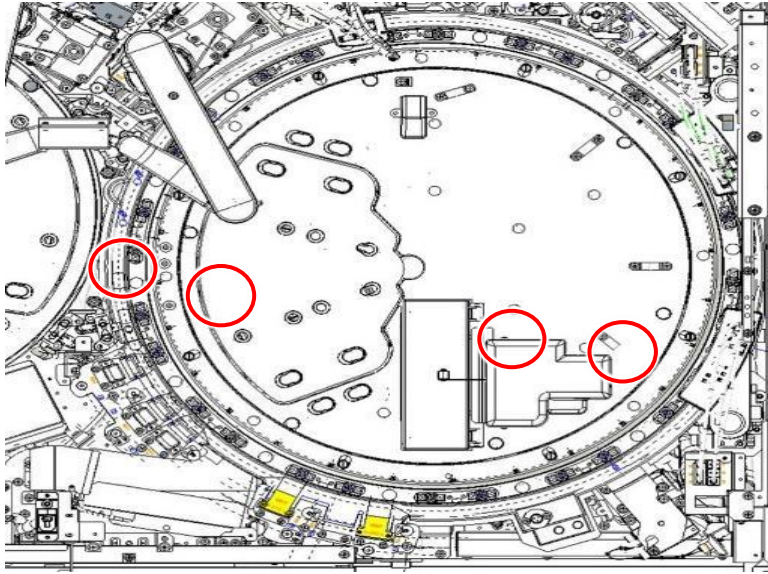


- (14) Rotate Reaction disk once counterclockwise and make sure that each arm (L) tip touches the touch point 1 to 4.
Note that the 2 arm (H) tips must not touch the touch points. If the 2 arm (H) tips touch the touch points, retry from the step (8).
If each arm (L) tip touches the touch point 1 to 4, proceed to the step (15).
If not, retry from the step (11).

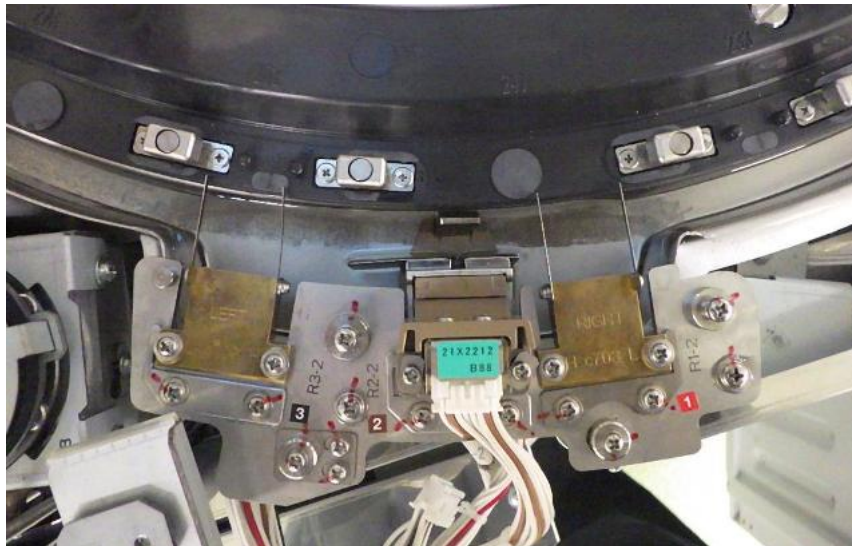


- (15) Make sure 2 arm (L) tips touch the same number of target points.
If 2 arm (L) tips touch the same number, proceed to the step (16).
If not, retry from the step (11).

(16) Loosen the 4 screws and remove the tools at the position of R1 and R3.



(17) Adjust the other USM units in the same way as step (5) to (16)

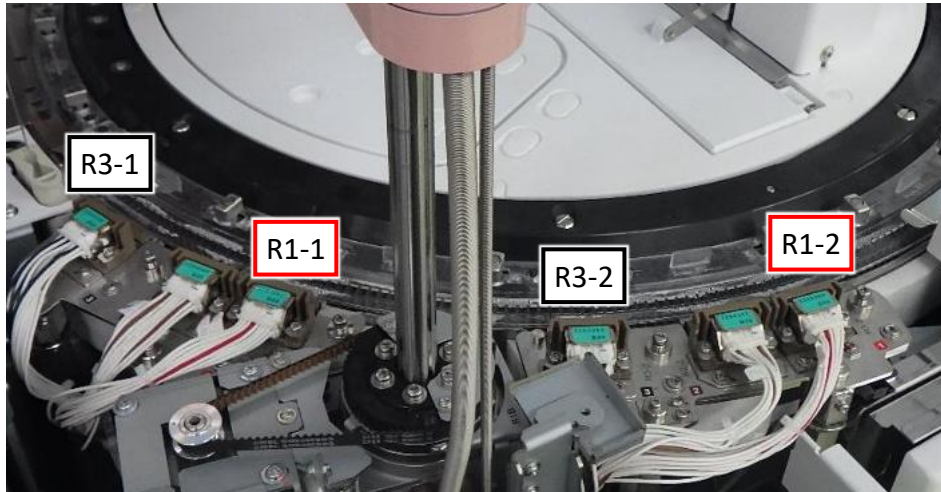


(18) After adjustment for USM units, attach the 4 USM units in their original positions.

(19) Attach the 14 cells.

(20) Refer to the following table and attach the 4 connectors.

Ultrasonic Mixing unit Position	R3-1	R1-1	R3-2	R1-2
Connector No.	P142	P140	P145	P143
Wire color	Black	Red	Black	red



(21) Select [End].

[System] - [Module] - [Service]

24/12/15V OFF

Adjustment

X-Y Z

Position

Adjusted

Current

Action

Accept

Back

Accept (Align)

Next

Set Up

Retry

BCR

Feed

L-Raised

Stop

End

848-3859

Touch points

Touch points

Locking Screw

Height Adjustment Bolt

Locking Nuts

Procedure

(a) Remove all cell segments and remove R1 and R3 USM units.

(b) Set the adjustment tool (PIN 848-3859) at the each USM position of R1 and R3.

(c) Loosen the Locking Nuts (x2) and Locking Screws (x2)

(d) Using the Height Adjustment Bolts (left and right), adjust so that the arms (L) occasionally touch the touch points of the reaction disk during manual rotation (less than 4 points) and the arms (H) keep no touching.

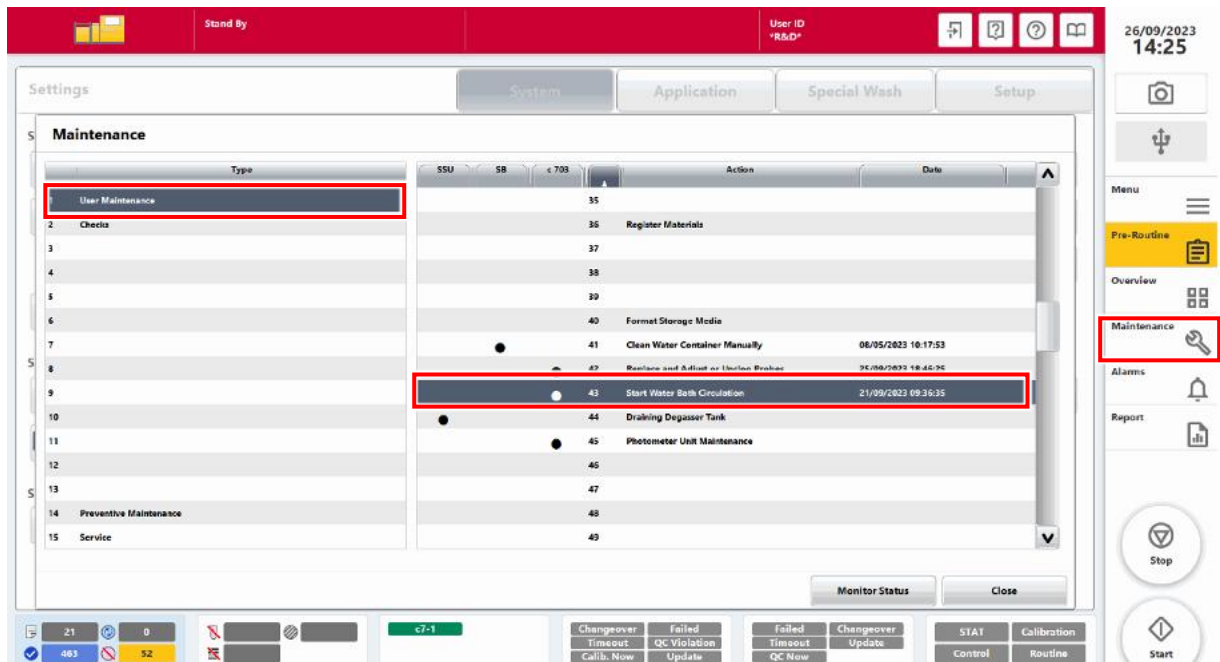
3.4 Notification

In case of the service software of “Ultrasonic Mixer Vertical” is terminated without selecting [End], perform the following steps after “Ultra sonic mixer Adjustment”.

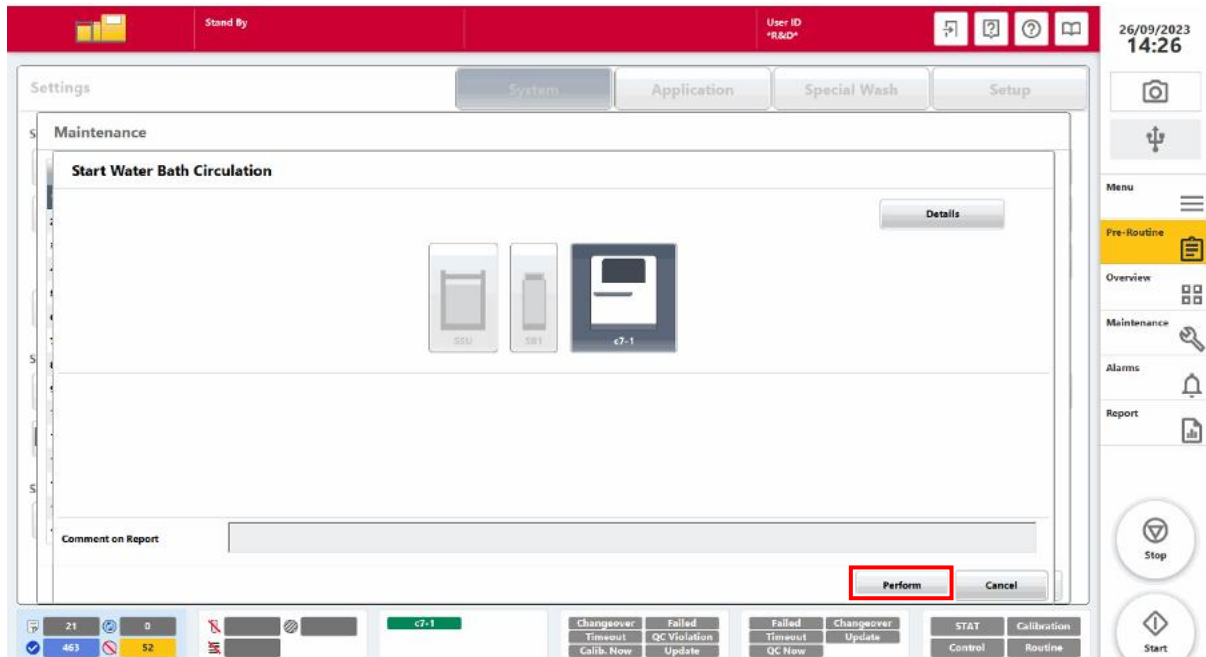
- (1) Turn off Maintenance switch.



- (2) Select [Maintenance] > [User Maintenance] > [Start Water Bath Circulation] on CU.

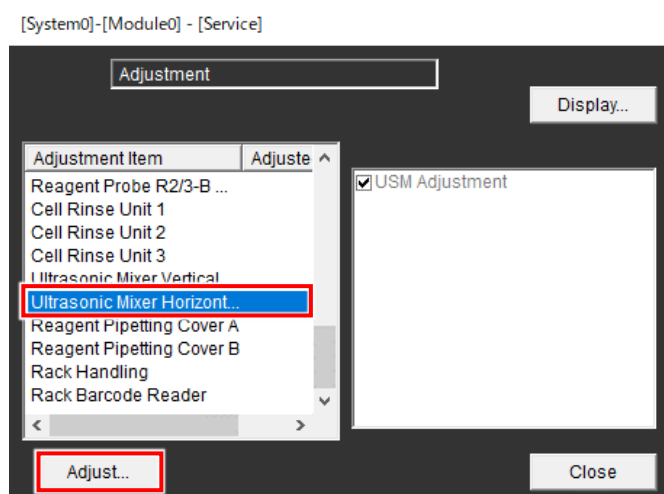


- (3) Select [Perform].

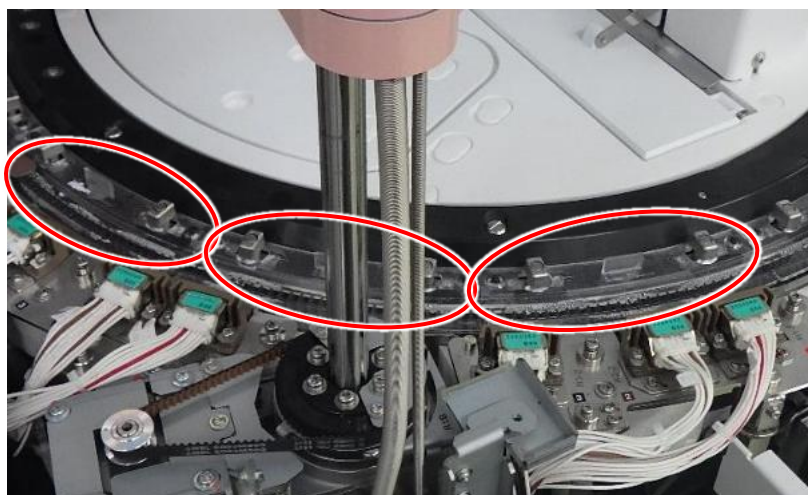


4. Adjustment of Ultrasonic Mixer Horizontal

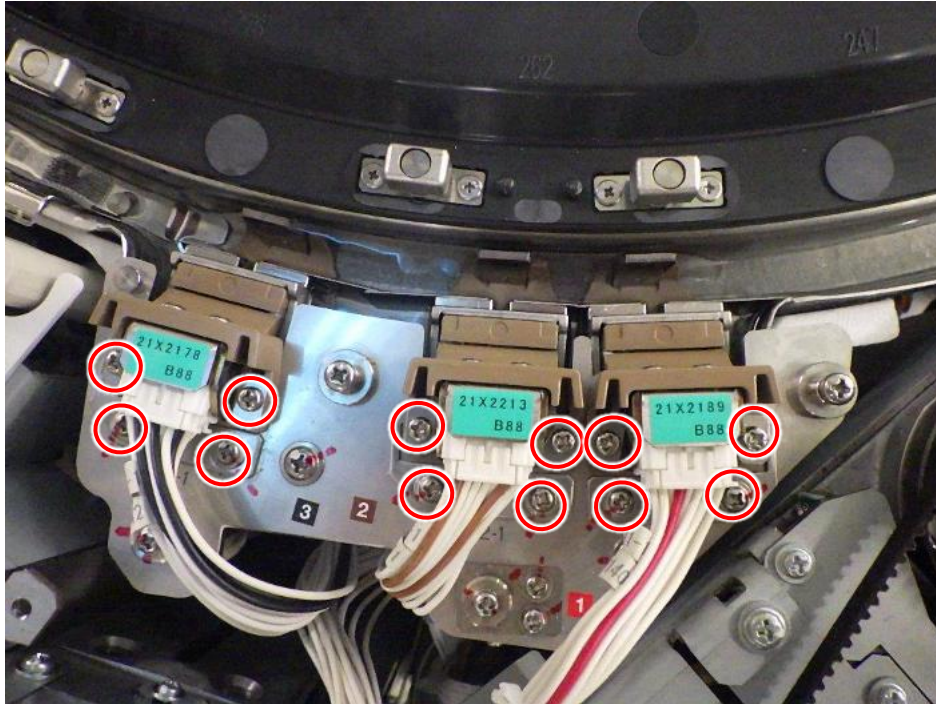
- (1) Select [Ultrasonic Mixer Horizontal] from the Adjustment Items and select [Adjust...].



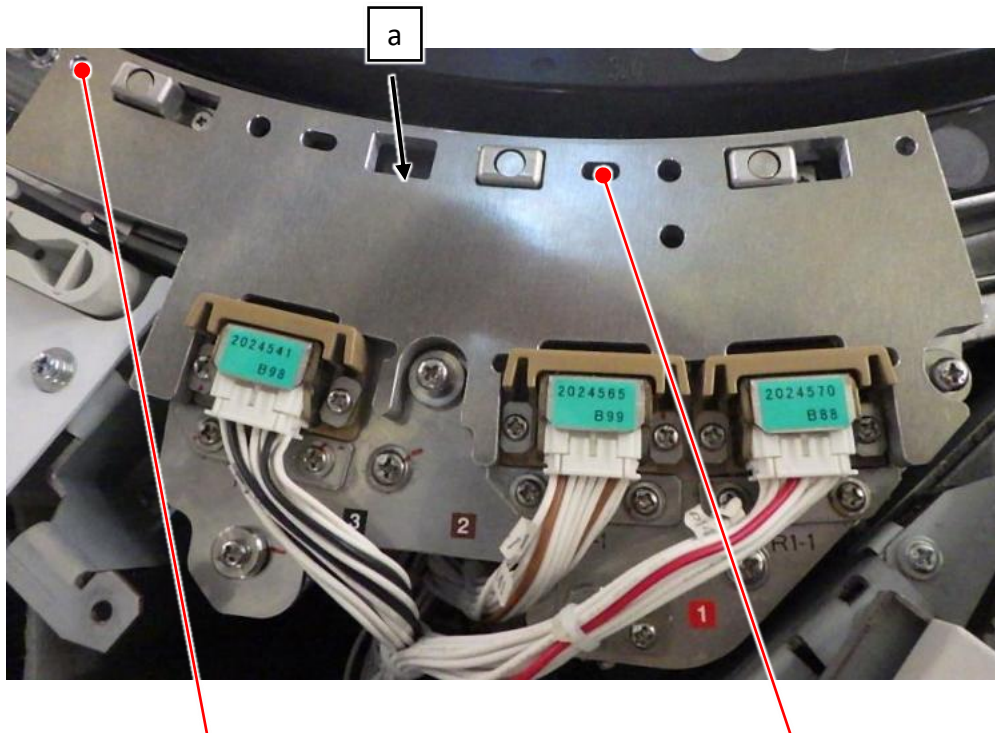
- (2) Remove the 3 cells in front of USM units.



- (3) Loosen the 12 screws.

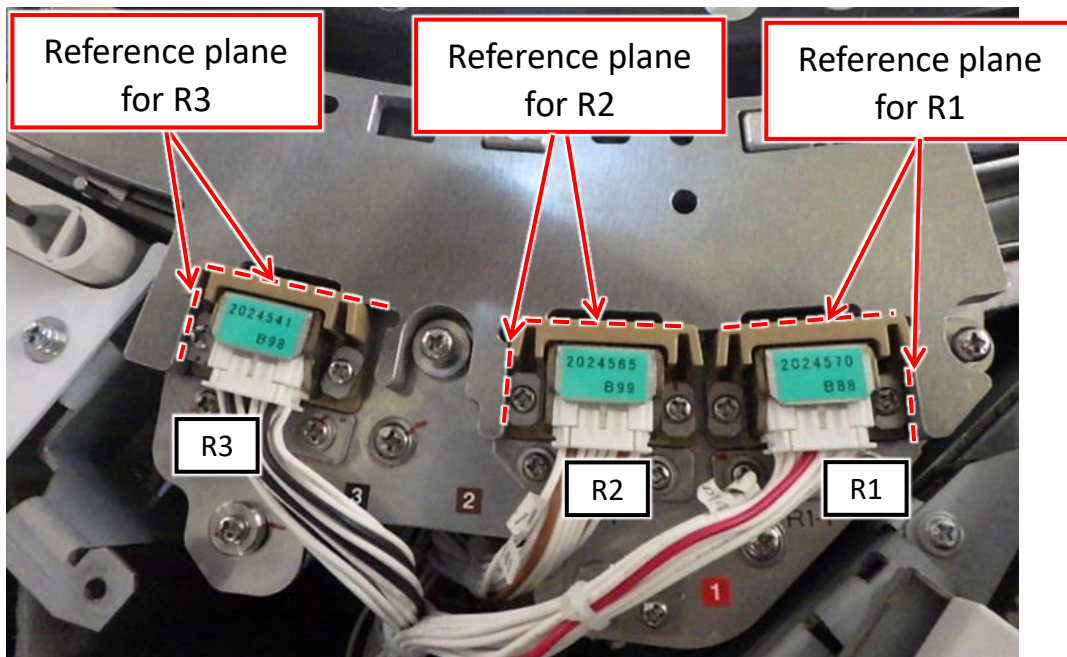


- (4) Set the tool (a) on Reaction disk to fit the two pins in the holes.

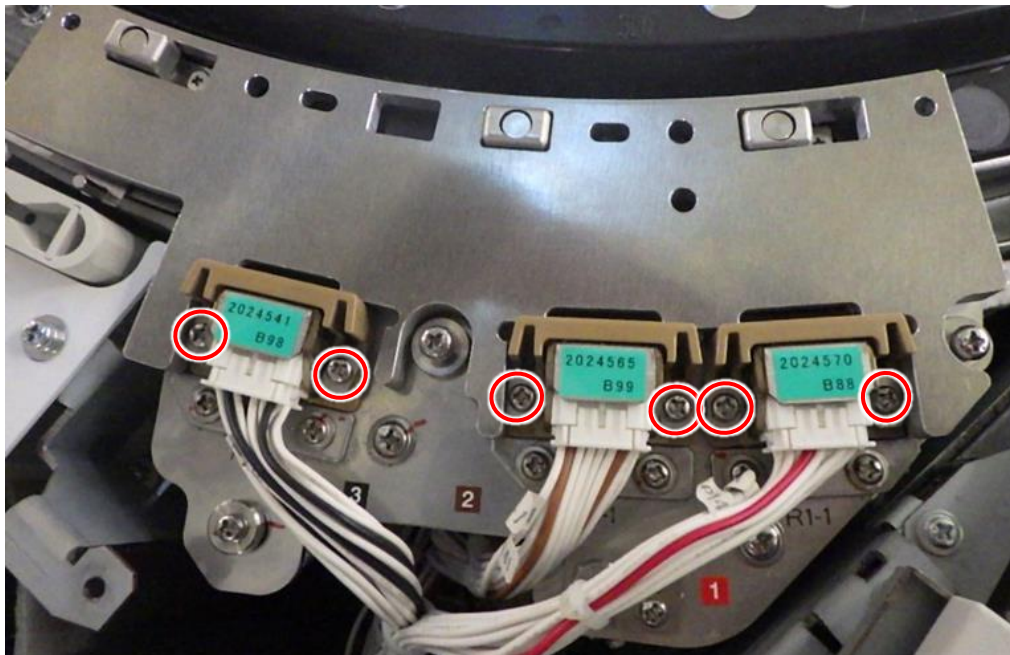


(a): USM TOOL (P/N: 848-3946)

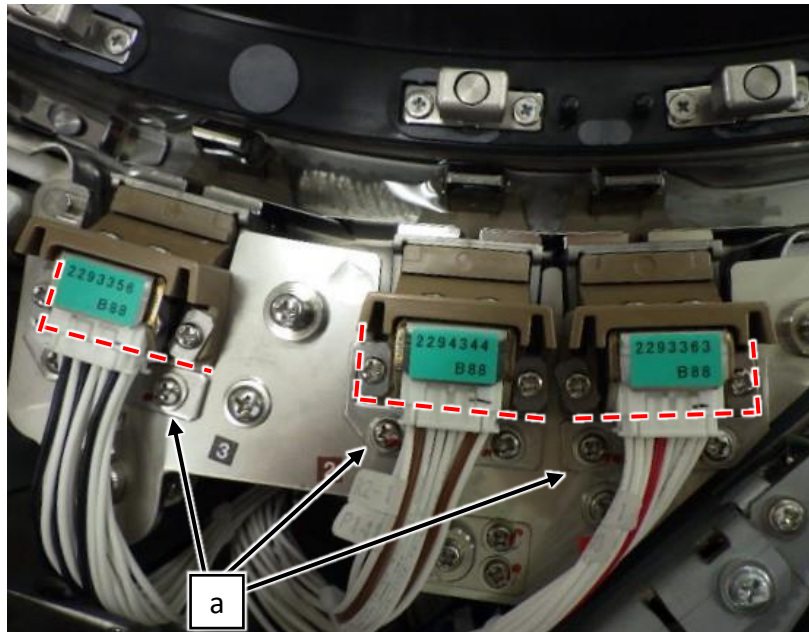
- (5) Fit each of 3 USM units to the 2 reference planes.



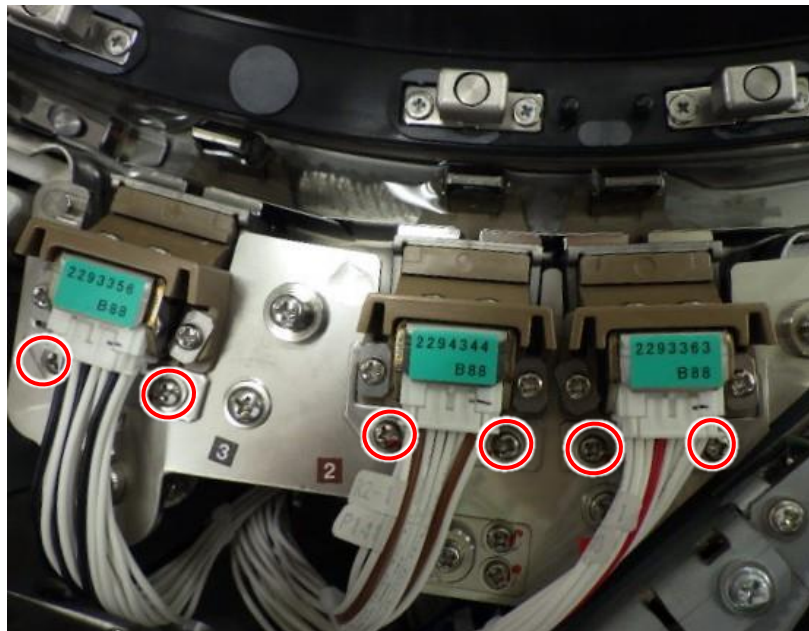
- (6) Tighten the 6 screws (a).



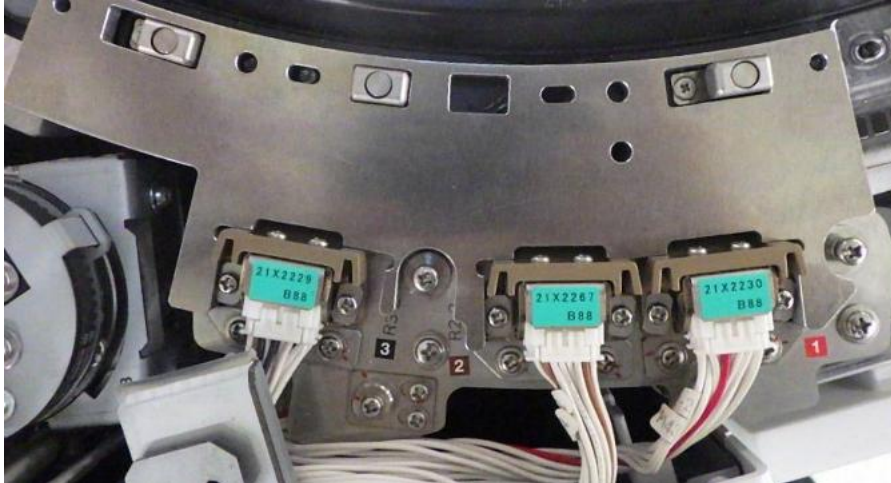
- (7) Remove the tool.
Note that the parts should be removed vertically.
- (8) Fit each of 3 plates (a) to the 2 sides of USM unit.



- (9) Tighten the 6 screws.



(10) Adjust the other USM units in the same way as step (2) to (9).



(11) Attach the 3 cells.

(12) Select [End]

[System0] - [Module0] - [Service]

USM Adjustment

Adjustment

X-Y

Z

Position

Adjusted

Current

Action

Accept

Accept & Next

Set Up

Back

Next

Retry

BCR

Feed

L.Reset

Stop

End

RO-1 positioning pin

RO-2 positioning pin

Reference plane R3

Reference plane R2

Positioning plate

Reference plane R1

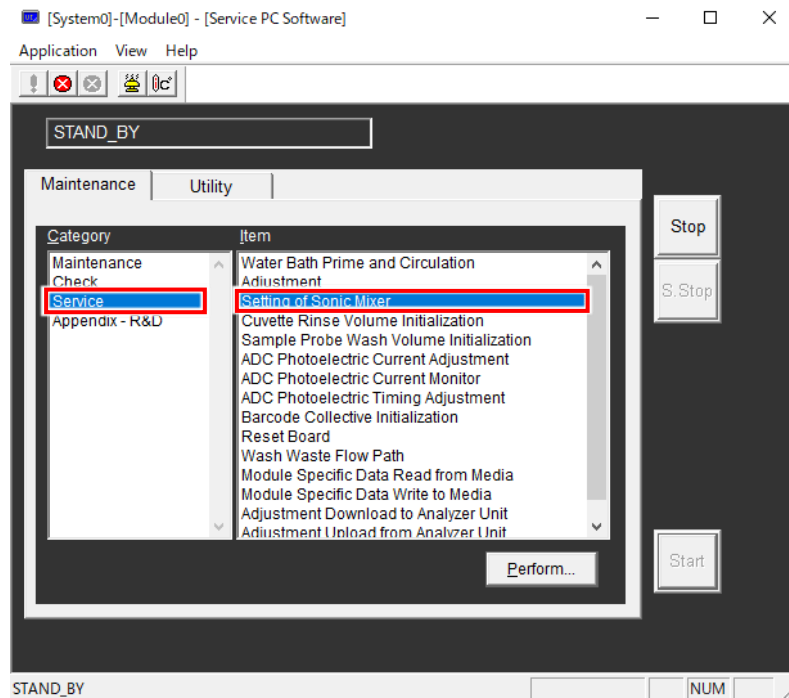
848-3197

Procedure

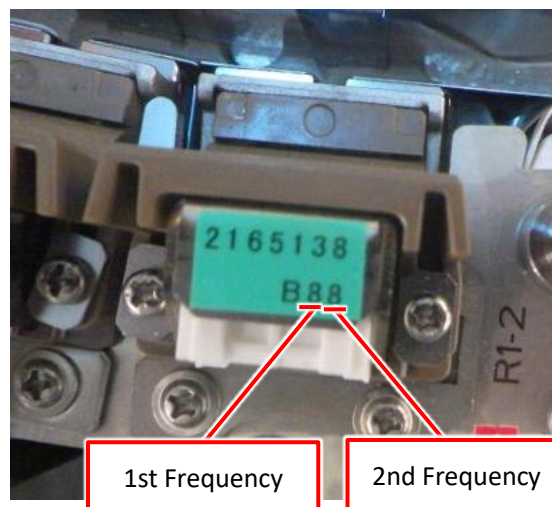
- Remove covers above USMs.
- Remove reaction cell segments around segments.
- Set the adjustment tool (P/N: 848-3197) for horizontal positioning on the cell positioning pins. If adjustment tool cannot be inserted into the correct position loosen positioning plates screws of USMs and USM screws and insert the adjustment tool.
- Move USM as shown on the diagram in direction of reference points (R1, R2 and R3), ensuring there is no gap between adjustment tool and USMs.
- Tighten USM screws in position.
- Move USM positioning plates so that there is no gap between USM and positioning plates.

5. Setting of sonic mixer

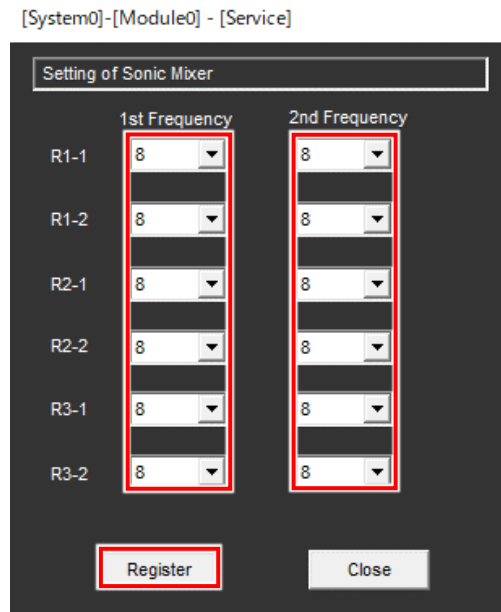
- (1) Select [Service] > [Setting of Sonic Mixer] > [Perform...].



- (2) Check the 1st and 2nd Frequency values of the R1 and R3 sonic mixers.



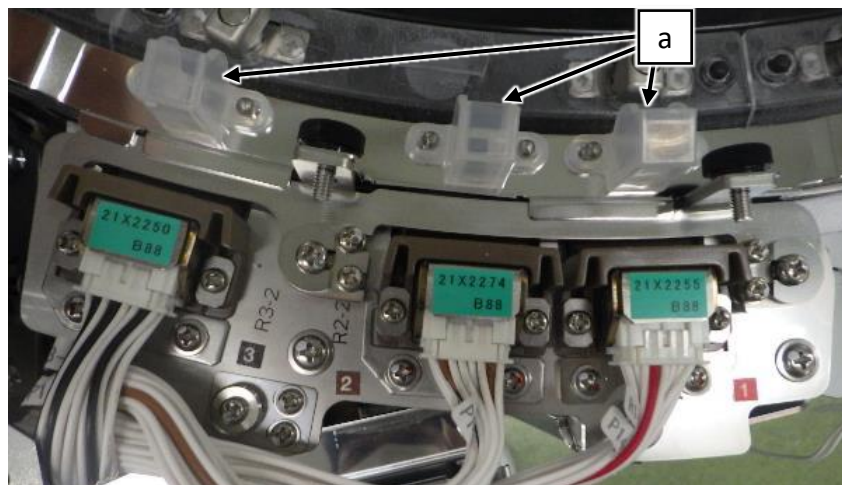
- (3) Set the 1st and 2nd Frequency values and select [Resister].



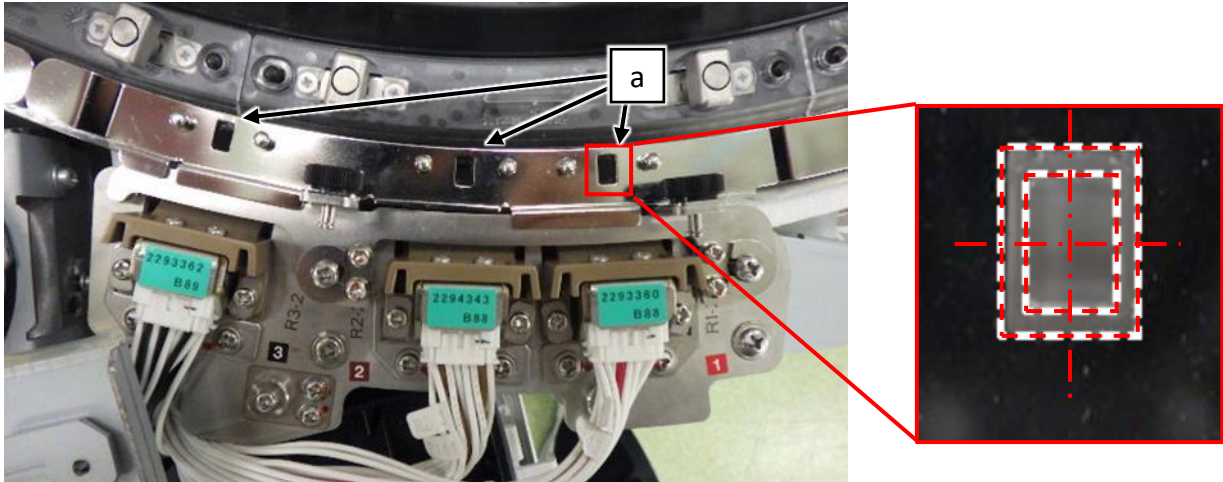
- (4) Select [Close].

6. Adjustment of USM cell cover ASSY

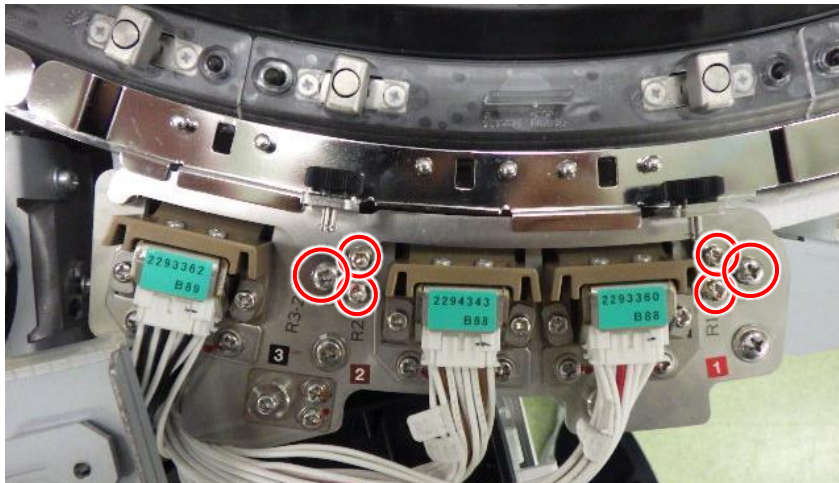
- (1) Attach USM cell cover ASSY.
(2) Remove the 3 splashing covers (a).



- (3) Make sure that the center of each hole (a) in USM cell cover ASSY lines up with the center of each cell.
If it lines up with the center, proceed to the next section.
If not, proceed to the next step.



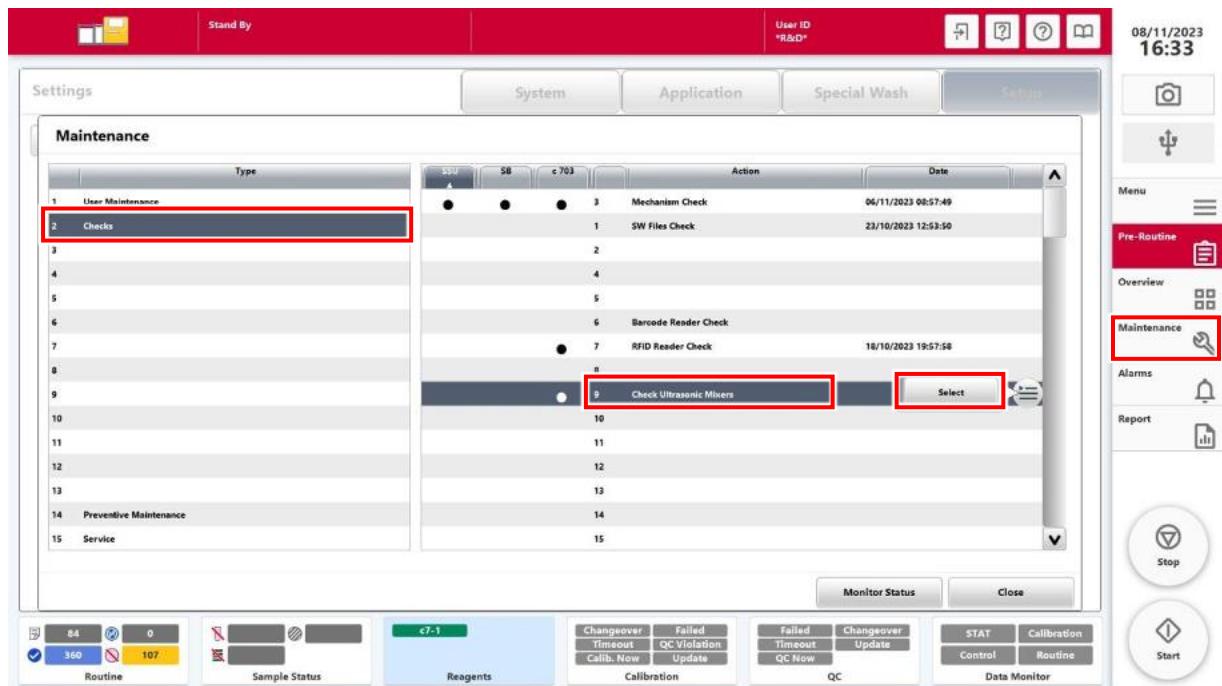
- (4) Loosen the 6 screws and adjust the position of USM cell cover ASSY so that the center of each hole in USM cell cover ASSY lines up with the center of each cell.



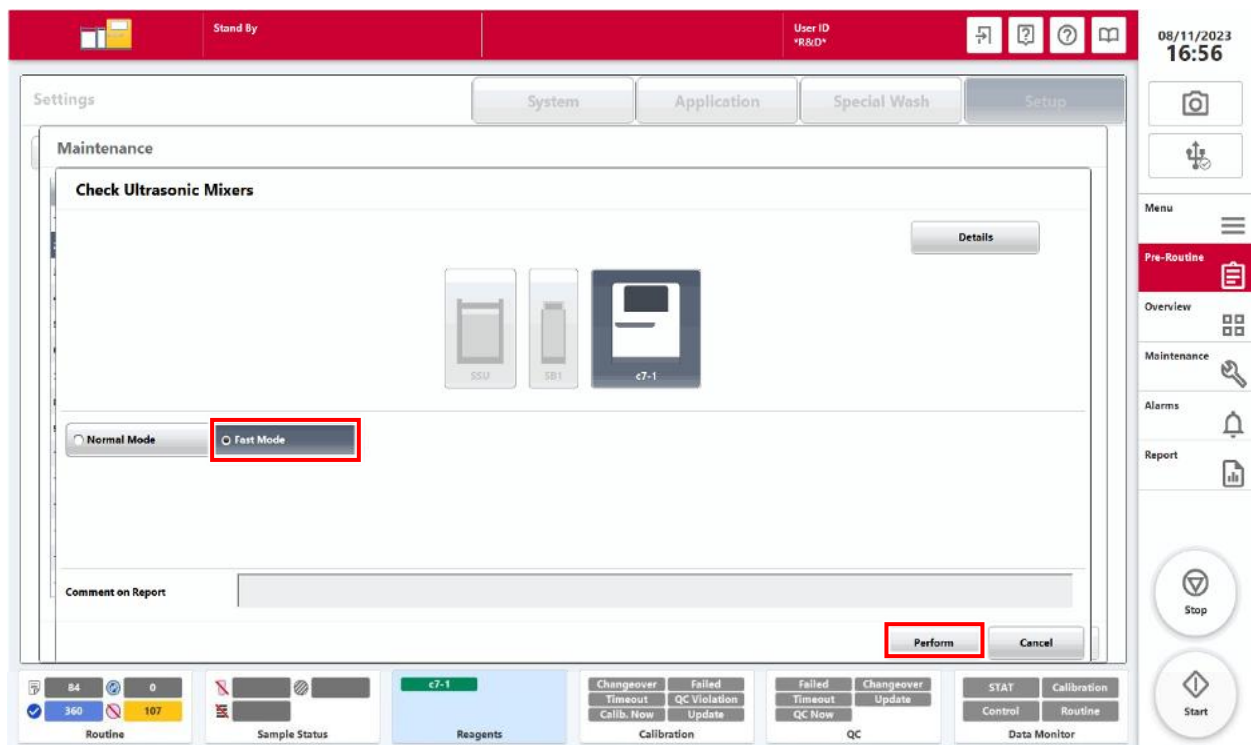
- (5) Tighten the 4 screws and attach the 3 splashing covers.
(6) Adjust the other USM cell cover ASSY in the same way as step (1) to (5).

7. Accomplishing the adjustment

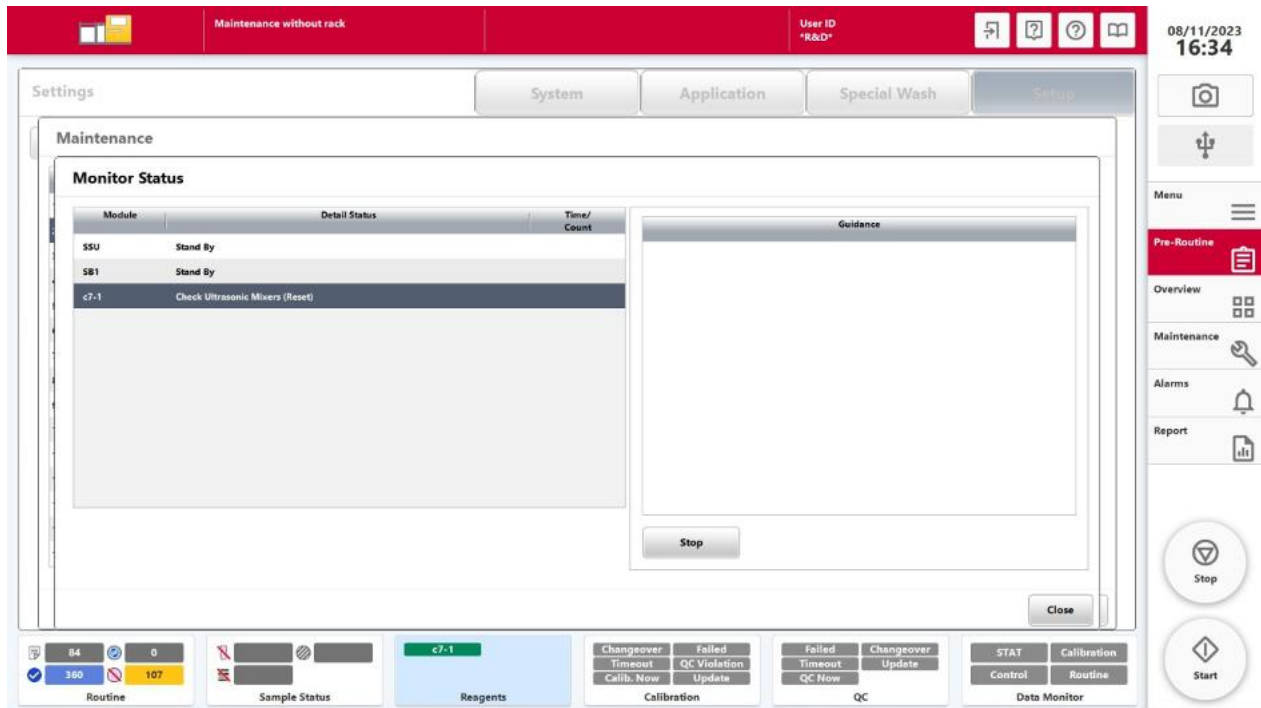
- (1) Turn off Maintenance switch.
- (2) Attach COVER MSW ASSY.
- (3) Attach No.1, No. 5, No. 6 and No. 13 covers.
- (4) Close COVER TOP F ASSY
- (5) Select [Maintenance] > [Check] > [Check Ultrasonic Mixers] > [Select] on CU.



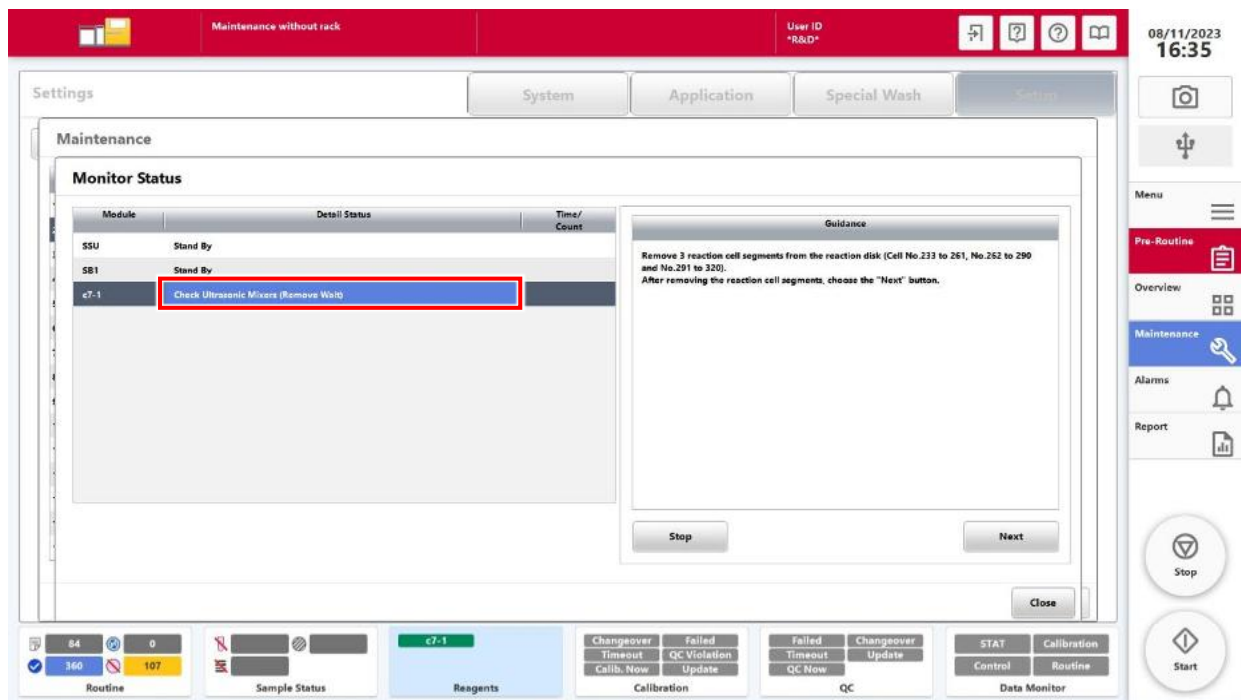
- (6) Select [Fast Mode] > [Perform].



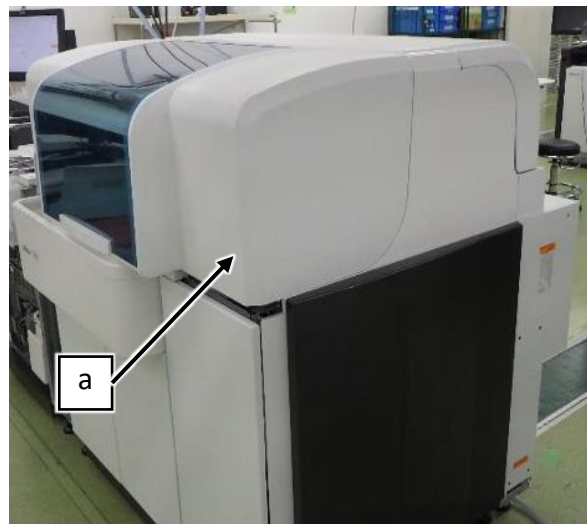
(7) Wait until the following screen appears.



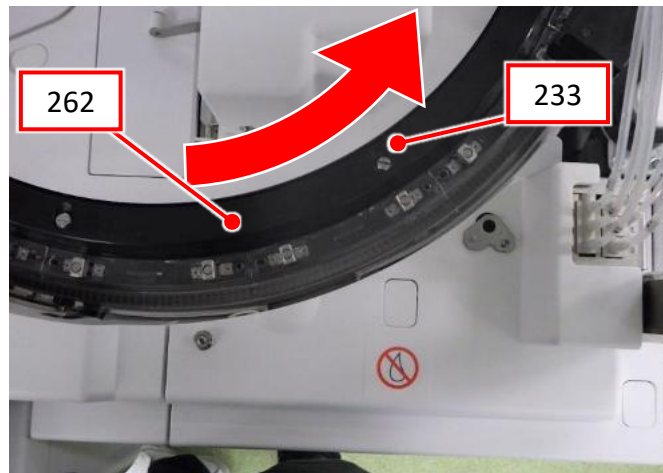
(8) Select [Check Ultrasonic Mixers (Return Wait)].



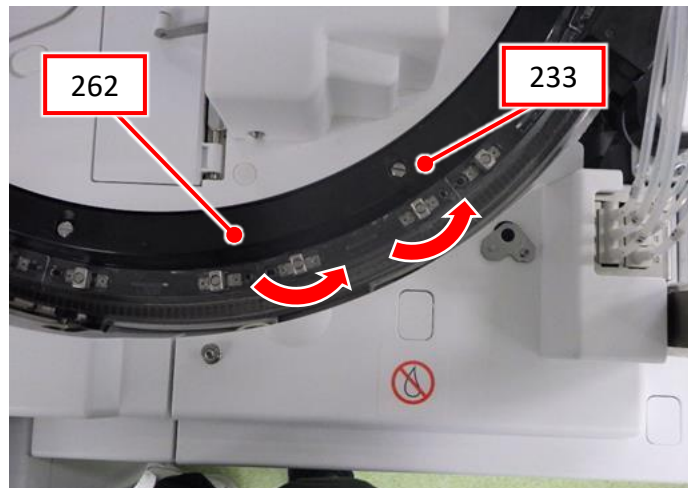
(9) Open the main cover (a).



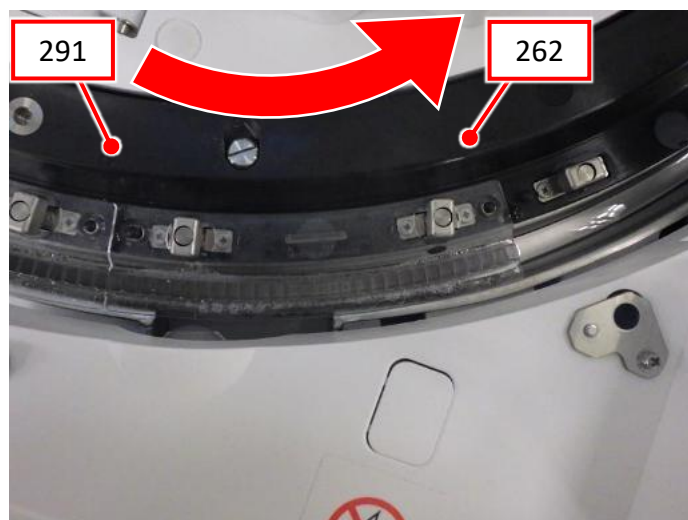
(10) Rotate Reaction disk counterclockwise until the cell numbers 233 and 262 at the positions as shown in the following figure.



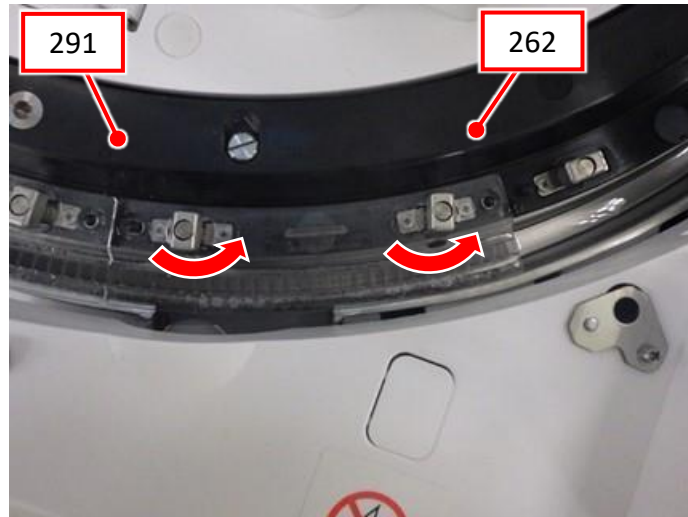
(11) Twist the 2 quick screws and remove the cell.



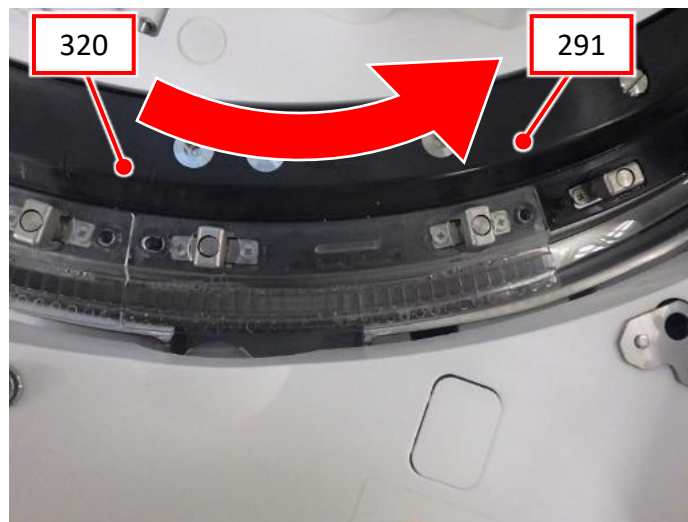
(12) Rotate Reaction disk counterclockwise until the cell numbers 262 and 291 at the positions as shown in the following figure.



(13) Twist the 2 quick screws and remove the cell.

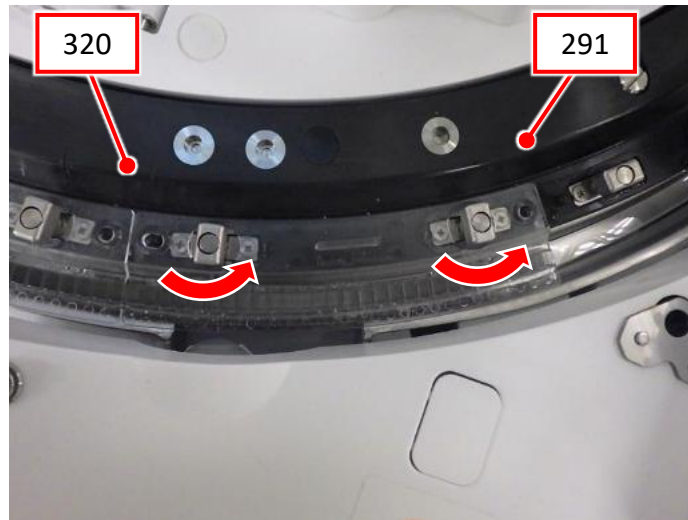


(14) Rotate Reaction disk counterclockwise until the cell numbers 291 and 320 at the positions as shown in the following figure.

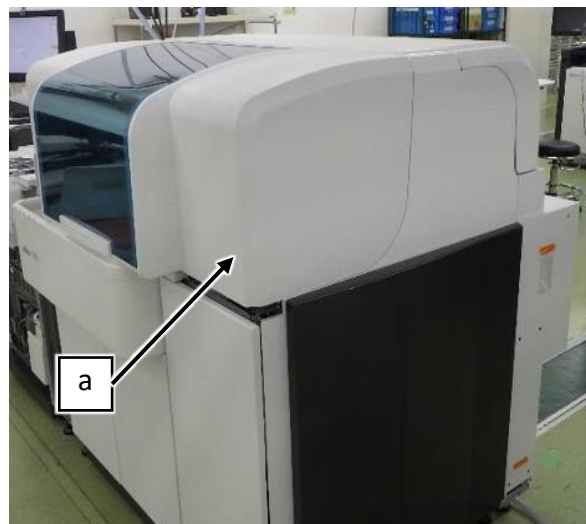


(15) Twist the 2 quick screws and remove the cell.

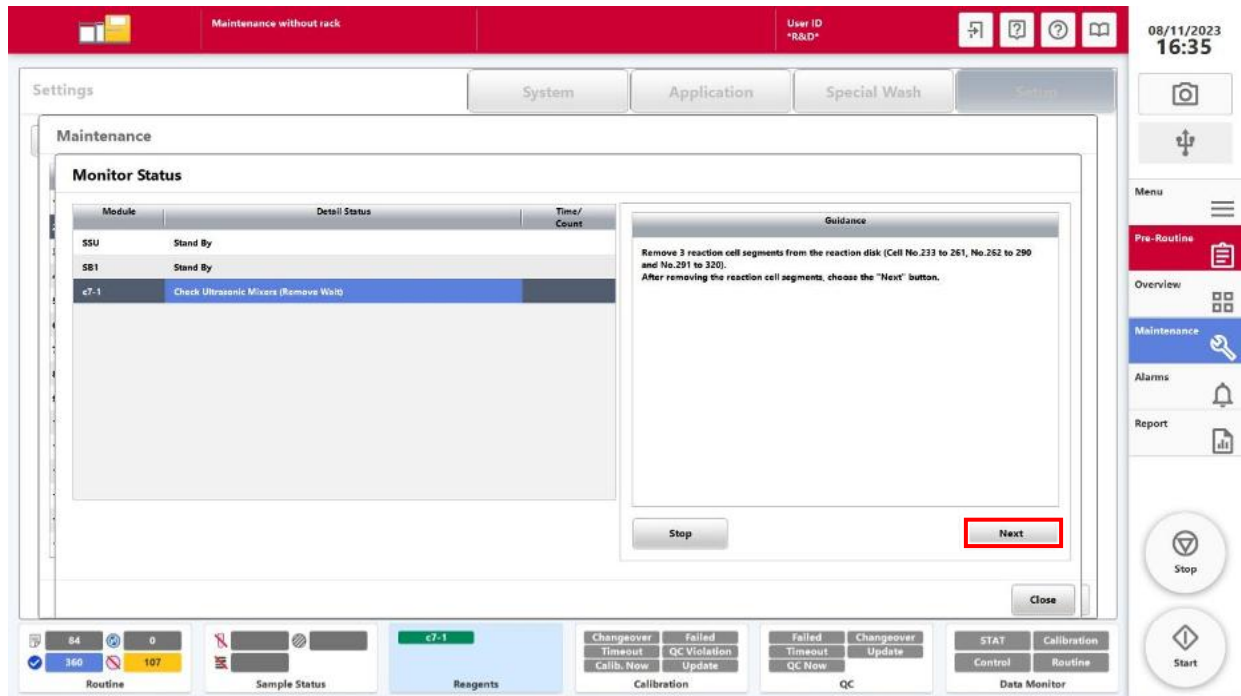
Note: Do not forget to remove the 3 cells. In case of forgetting, USM units may break.



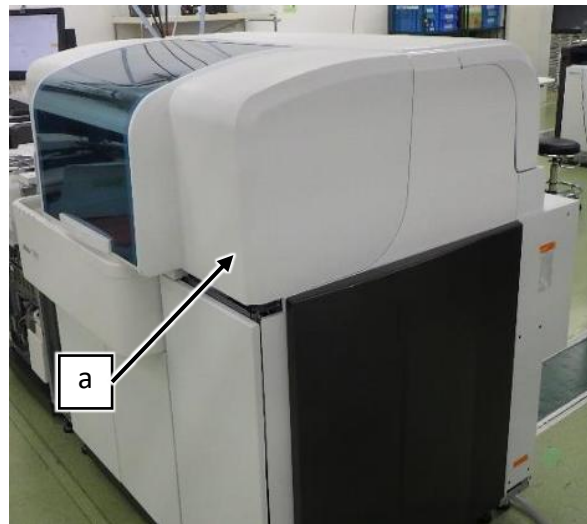
(16) Close the main cover (a).



(17) Select [Next].



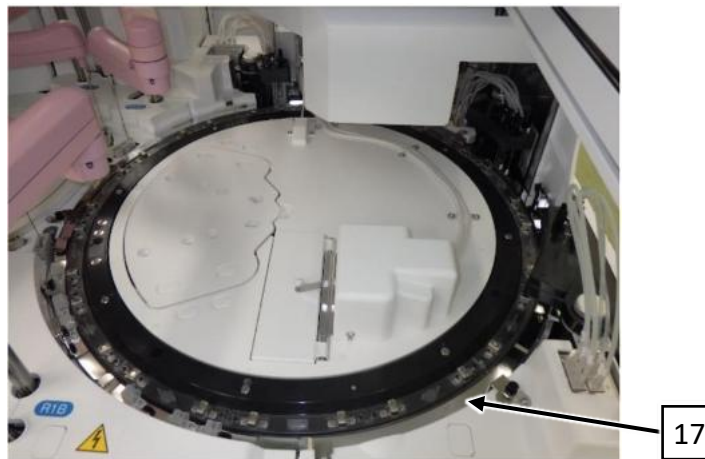
(18) After Check Ultrasonic Mixers, open the main cover (a).



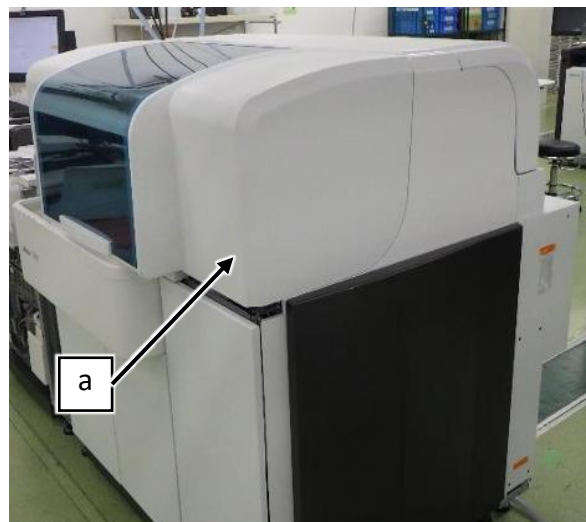
(19) Attach the 3 cells and twist each 2 quick screws.



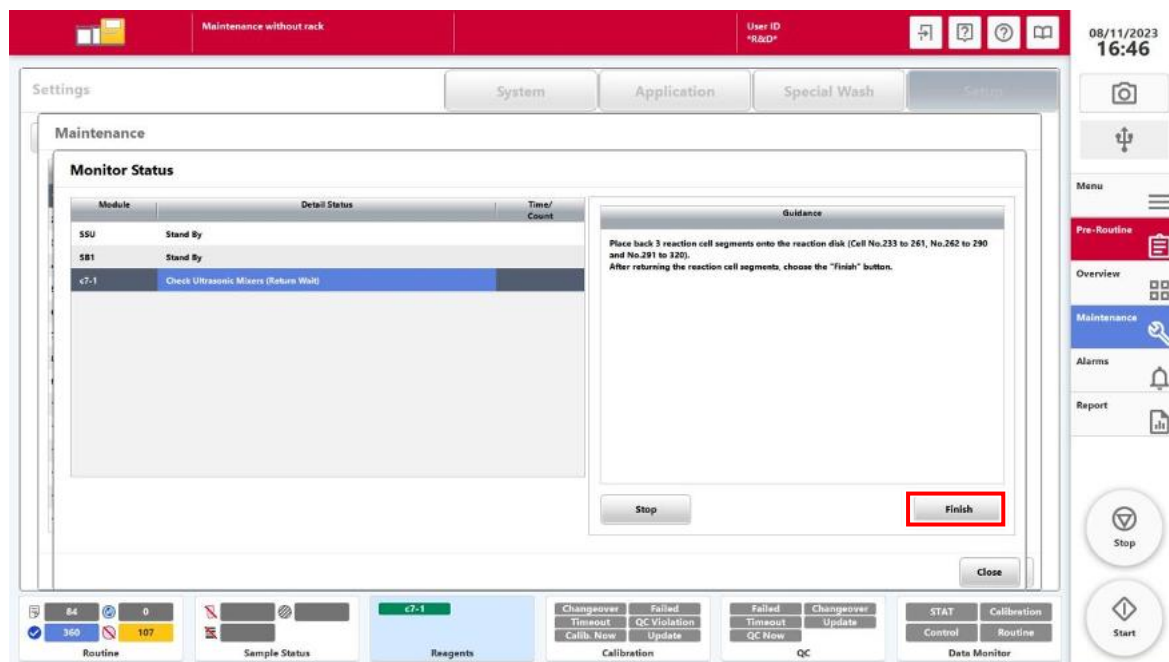
(20) Attach No. 17 cell cover (refer to “S06-41 Removal and attachment of covers”).



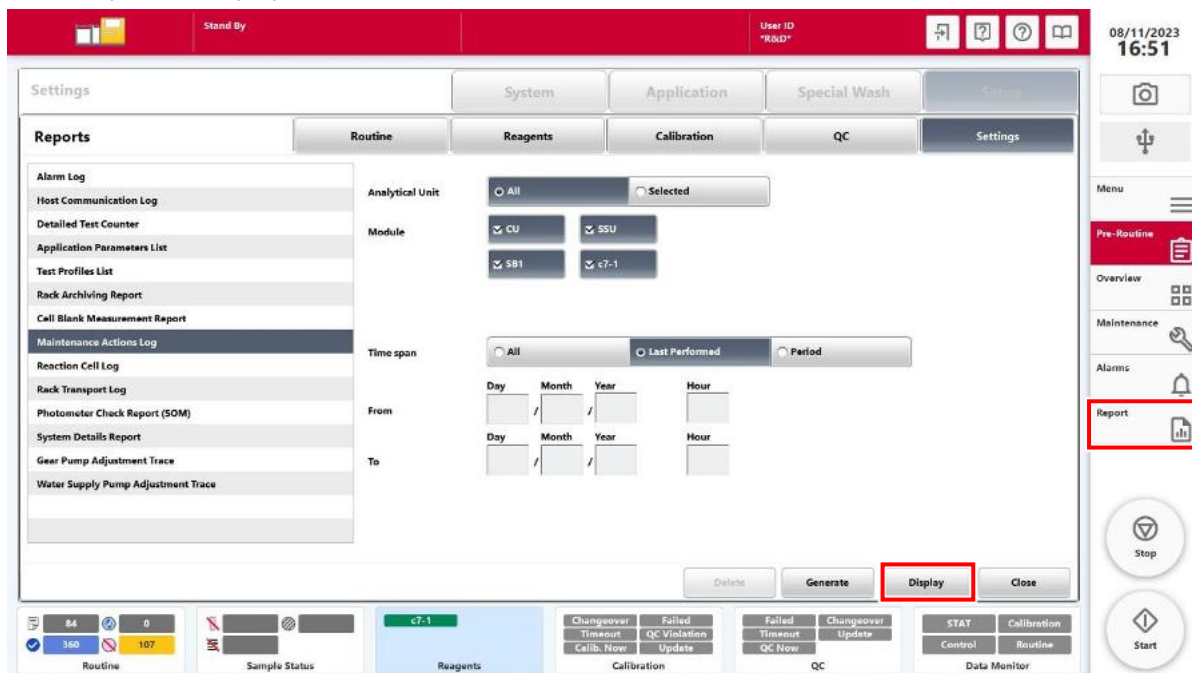
(21) Close TOP COVER F ASSY (a).



(22) Select [Finish].



(23) Select [Report] > [Display].



(24) Select [Check Ultrasonic Mixers].

The screenshot shows the Hitachi Cobas c703 service menu. The 'Display' list on the left has 'Check Ultrasonic Mixers' highlighted. The main display area shows the 'Check Ultrasonic Mixers' screen with a table of data for three mixers (R1, R2, R3). The table includes columns for Date/Time, Mode, Mixing Step, Frequency, Limit, and Pattern1. The 'Current' column for each mixer shows 'OK' (✓) for all three mixers. The 'Previous' column shows 'NG' (✗) for all three mixers. The 'Limit' column shows '180' and '600' for each mixer. The 'Pattern1' column shows '58', '57', and '53' for mixers R1, R2, and R3 respectively. The 'Limit' column shows '117', '111', and '111' for mixers R1, R2, and R3 respectively. The 'Pattern1' column shows '118', '112', and '111' for mixers R1, R2, and R3 respectively. The 'Limit' column shows '120' for mixer R1. The 'Pattern1' column shows '111' for mixer R3. The 'Limit' column shows '111' for mixer R3. The 'Pattern1' column shows '111' for mixer R3.

(25) Make sure that all symbols of “Current” are OK.

If not, check the setting of sonic mixer (refer to the chapter 5) and each plug is inserted (refer to the step 3.3 (20)).

The screenshot shows the Hitachi Cobas c703 service menu. The 'Check Ultrasonic Mixers' screen is displayed. The 'Current' column for each mixer (R1, R2, R3) is highlighted with a red box, showing 'OK' (✓) for all three mixers. The 'Previous' column for each mixer shows 'NG' (✗) for all three mixers. The 'Limit' column shows '180' and '600' for each mixer. The 'Pattern1' column shows '58', '57', and '53' for mixers R1, R2, and R3 respectively. The 'Limit' column shows '117', '111', and '111' for mixers R1, R2, and R3 respectively. The 'Pattern1' column shows '118', '112', and '111' for mixers R1, R2, and R3 respectively. The 'Limit' column shows '120' for mixer R1. The 'Pattern1' column shows '111' for mixer R3. The 'Limit' column shows '111' for mixer R3. The 'Pattern1' column shows '111' for mixer R3.

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